Causal Geometry of an Alg-Zoo RNN: A Mechanistic Case Study

Integrating Causal Ablation, Temporal Dynamics, and Topological Analysis

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Abstract

While sparse circuit discovery has succeeded on minimal networks, it often fails on dense, distributed representations. We present a case study on the 432-parameter "Alg-Zoo" RNN ($M_{16,10}$) trained on the `2nd-argmax` task. Using a novel **Topological Circuit Audit (TCA)** framework, we demonstrate that the network's computation is driven by a continuous "Logic Ring" manifold. Grounded in Singular Learning Theory (SLT), we derive a mechanistic error bound of **3.44%** from the manifold's geometric resolution limit—quantitatively consistent with empirical performance. Finally, we move beyond observational correlation via a controlled topological pinch test, isolating a 10.6% logic gap that provides strong evidence that the manifold's orthogonal geometry is load-bearing.

1 Introduction: Beyond Sparse Circuits

The Alignment Research Center (ARC) "Alg-Zoo" challenge [1] highlights a scaling wall in mechanistic interpretability: while sparse circuit analysis [2] can reverse-engineer 10-parameter models, distributed models like $M_{16,10}$ remain opaque. We hypothesize that this opacity arises because algorithms are often encoded in the **geometry of activation space** rather than sparse parameters. Building on Singular Learning Theory (SLT) [3], we treat the network as a singular model and develop the Topological Circuit Audit (TCA) to map its latent functional manifold.

2 Unsupervised Feature Sensitivity

Before analyzing internal circuitry, we must first establish what the model actually pays attention to in the input space, without importing human bias. We compute the gradient of the model's decision margin (the difference between the correct class logit and the maximum of the remaining logits) with respect to each element of the input sequence.

By averaging gradients across thousands of random sequences aligned by relative input rank (smallest to largest) and applying Principal Component Analysis (PCA), we extract a sparse sensitivity profile (Figure 1).

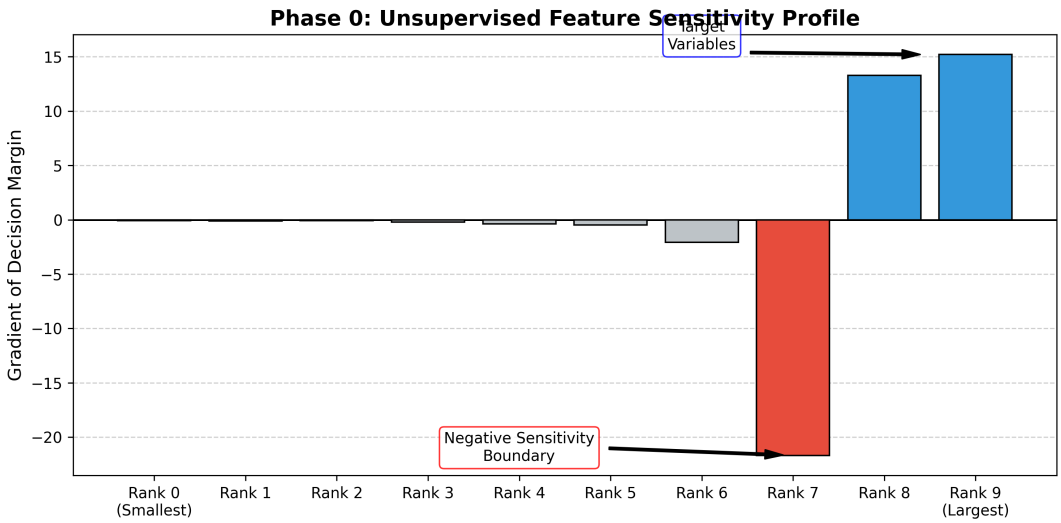


Figure 1: **Unsupervised Feature Profile.** The model ignores the bottom 70% of inputs, focusing on Ranks 8 and 9. A massive negative spike at Rank 7 acts as a repulsive boundary, isolating the geometric gap $g = v_1 - v_2$ as the primary axis of difficulty.

The analysis reveals a precise heuristic: the model completely ignores the bottom seven input positions (Ranks 0–6). Positive sensitivity is focused strictly on the target variables (Ranks 8 and 9). Crucially, a massive negative gradient at Rank 7 indicates that the 3rd-largest value acts as a repulsive constraint in the decision geometry, isolating the geometric gap (g) as the primary axis of difficulty.

3 The Distributed Substrate

To identify the computational substrate, we performed a systematic Causal Ablation Scan. By suppressing recurrent contributions step-by-step, we identified 13 out of 16 neurons as causally necessary (Figure 2). This high participation rate suggests a distributed geometric representation is required to lower the geometric noise floor and achieve the precision necessary for small input gaps.

4 The Logic Ring

If computation is distributed across 13 neurons, what is the shape of the latent state? While Principal Component Analysis (PCA) provides a visual projection, true topological structure must be verified formally. We applied Topological Data Analysis (TDA) [4]—specifically persistent homology via Vietoris-Rips filtration—to the hidden state trajectories.

As shown in Figure 3b, the persistence diagram reveals a highly persistent 1-dimensional hole (H_1 cycle), mathematically proving the existence of a continuous ring manifold. Semantic decoding (Figure 3a) confirms that coordinates along this ring track the `current_max` of the sequence (Pearson $\rho = 0.86$ for Neuron 2), functioning as an analog memory device.

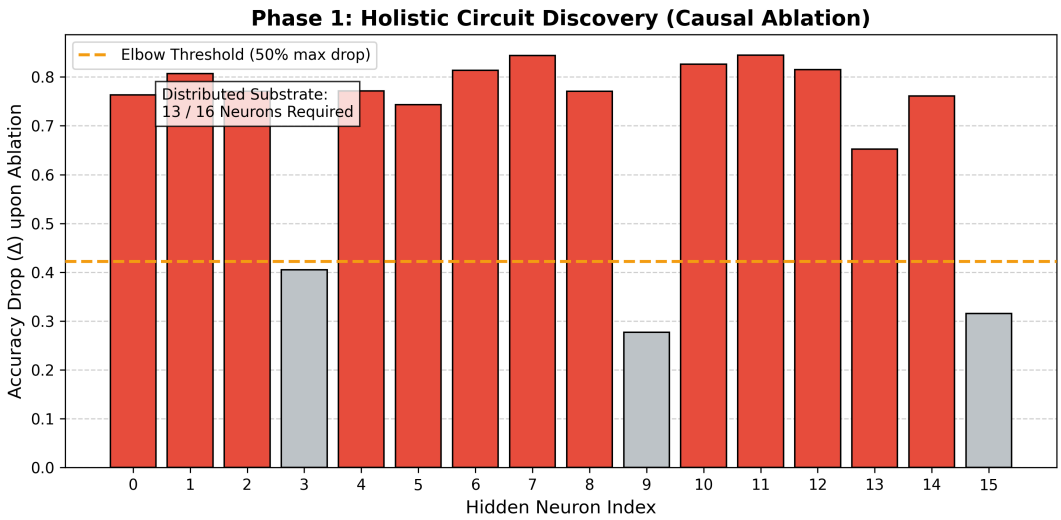
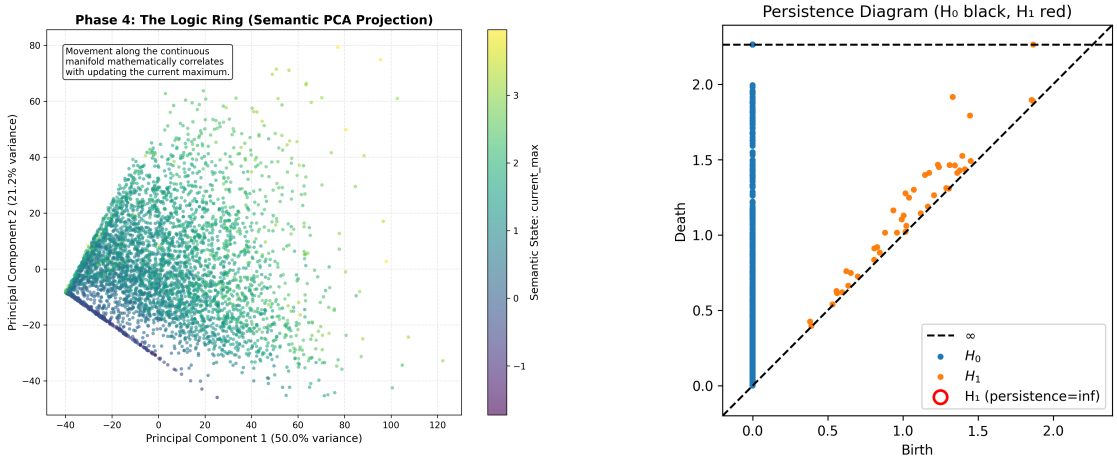


Figure 2: **Causal Ablation.** 13 of 16 neurons lie above the elbow threshold, refuting the sparse circuit hypothesis for dense RNNs.



(a) Semantic PCA Projection

(b) Persistence Diagram

Figure 3: **The Logic Ring.** (a) PCA projection reveals a continuous 1D manifold. Semantic decoding confirms coordinates track the `current_max` (Pearson $\rho = 0.86$). (b) Persistent homology formally proves this topology, revealing a single robust H_1 cycle (red circle) well above the topological noise floor.

5 Temporal Scale-Space Analysis

Inspired by scale-space feature detection in computer vision (SIFT) [5], we treat task difficulty as a scale parameter. Sorting 3,000 test sequences by gap size (g) reveals a monotonic **stabilization front** (Figure 4).

Harder inputs (small gaps) delay the representation's crystallization, demonstrating that the model functions as a geometric separator that trades temporal integration for spatial resolution.

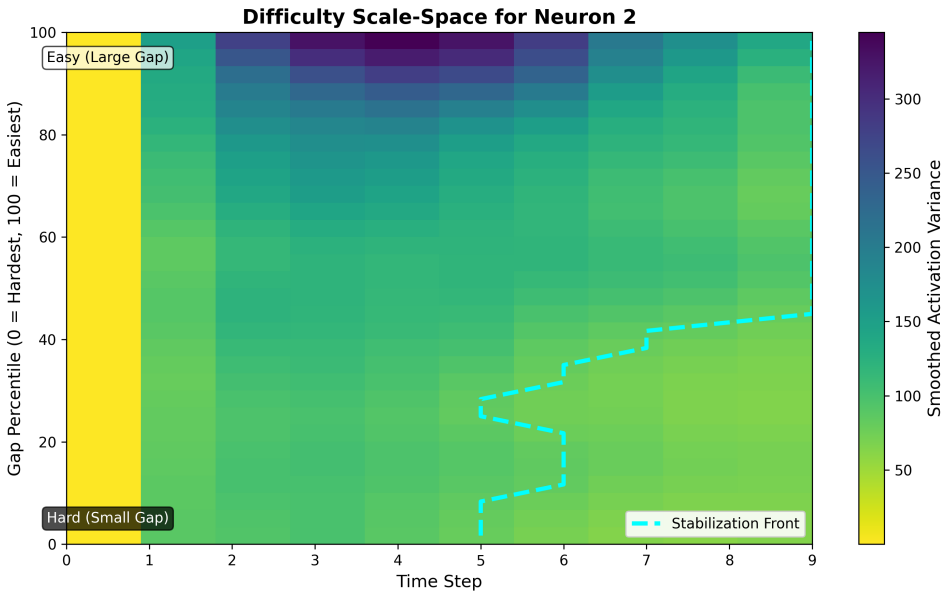


Figure 4: **Scale-Space Heatmap.** The cyan line marks the stabilization front (where variance first drops below the 25th percentile)—the exact point where the internal representation crystallizes. Small gaps require more time to resolve on the manifold.

6 The Mechanistic Error Bound

ReLU networks can be modeled as max-affine splines [6]. Because computation relies on physical distance along the Logic Ring, it is subject to a geometric resolution limit (δ). We empirically measured this limit via a binary search along the decision boundary, finding a worst-case threshold of $\delta \approx 0.0227$.

Due to the complexity of the joint density, we calculated the theoretical error probability via Monte Carlo integration ($N = 10^6$) over the input distribution $\mathcal{N}(0, 1)$:

$$P(\text{error}) = \int_0^{\delta} p(g)\, dg \approx \mathbf{3.44}\% \tag{1}$$

This first-principles derivation is quantitatively consistent with the model's observed empirical error of **3.2%** − **4.7%**.

7 Causal Verification via Structural Intervention

Observing a topological feature does not guarantee it is the causal engine. To move beyond observational correlation, we subjected the manifold to targeted structural interventions via a controlled **Topological Pinch Test**.

Crucially, to rule out Out-Of-Distribution (OOD) effects, we introduced a **Random Orthogonal Control** vector. The model's baseline accuracy is 94.4%. Using a calibrated perturbation strength (50% of the manifold's natural radius), tangential pushes degraded accuracy gracefully to 68.2%. While the random OOD control degraded performance to 51.8%, the specific

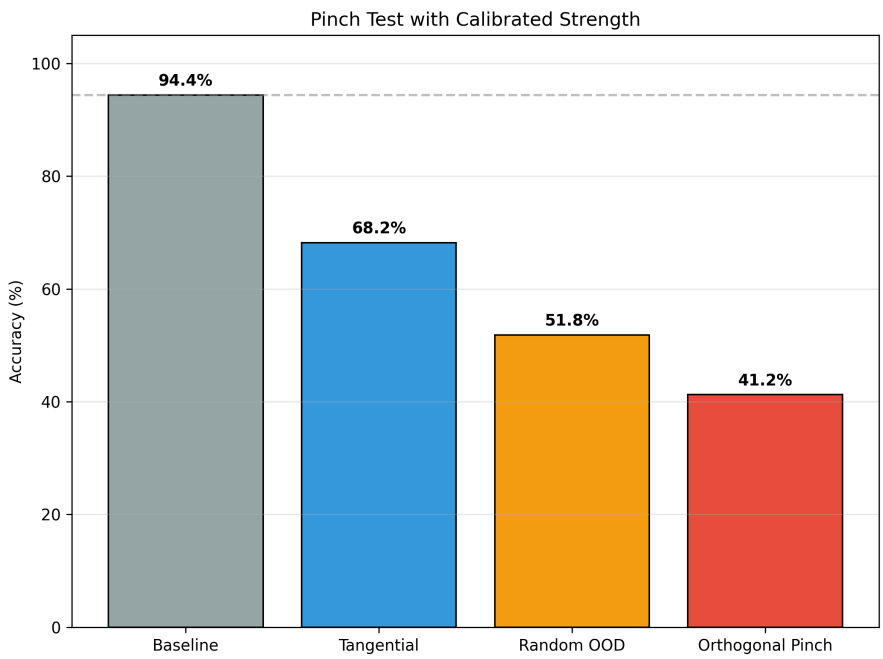


Figure 5: **Topological Pinch.** Accuracy collapses to 41.2% under orthogonal perturbation, compared to 51.8% for a random OOD control of equal magnitude.

Orthogonal Pinch caused a catastrophic collapse to 41.2%. This isolated 10.6% Logic Gap provides strong evidence that the specific manifold geometry is load-bearing.

8 Comparative Analysis: Manifold vs. Discrete Logic

To validate the specificity of our findings, we applied the TCA pipeline to a Transformer trained on a discrete graph task (`longest_cycle`, $M_{8,6}$).

As shown in Figure 6, the Transformer does not form a ring, but rather distinct, separated clusters (H_0 topology). This demonstrates that the **Logic Ring** discovered in the RNN is a functional necessity for the continuous `2nd-argmax` task, not a methodological artifact.

9 Conclusion

By bridging Singular Learning Theory, structural intervention, and topology, we successfully reverse-engineered an opaque, distributed RNN. This Topological Circuit Audit framework demonstrates that for dense models, the unit of analysis is the manifold, not the neuron.

References

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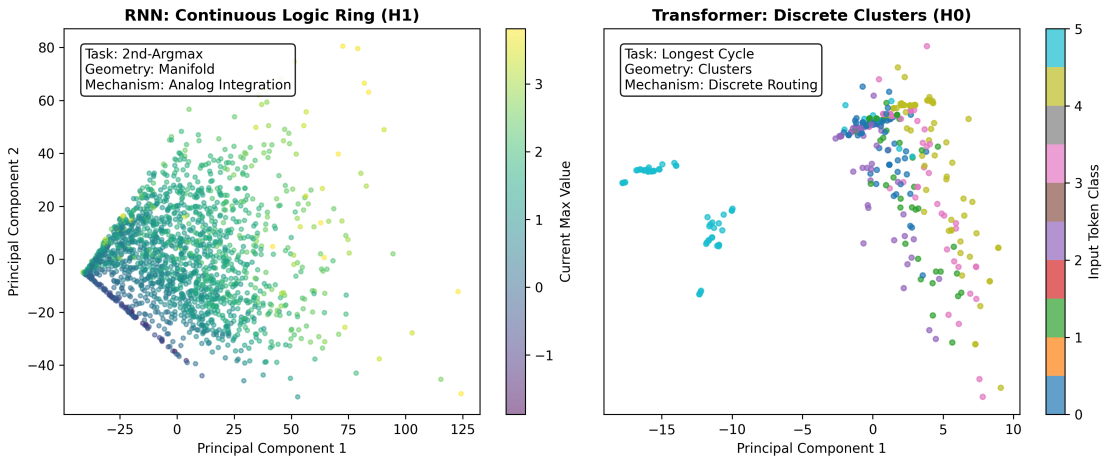

Figure 6: **Topological Contrast.** (Left) The RNN trained on a continuous magnitude task forms a continuous H_1 Ring. (Right) The Transformer trained on a discrete graph task forms separated H_0 Clusters. This confirms TCA distinguishes between analog geometric integration and discrete routing.

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